

Supplementary Appendix to “Modelling New Zealand Student Visa Approval Rates with Country-Level Indicators”

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The appendix tables refer to repository-relative files in the project repository [1]. The source and current PDF of this supplementary appendix are stored in the repository’s `appendix/` directory.

A Repository Structure

The repository separates source data, prepared data, model results, and support outputs so that the empirical workflow can be inspected without mixing raw inputs with derived files. All paths mentioned in this article are relative to the repository root.

`data/original/` contains source files downloaded from INZ and from reference-data providers. These files preserve the input record and are not edited manually.

`data/prepared/` contains cleaned, harmonised, aggregated, and joined datasets. These files are used as intermediate research tables and as modelling inputs after country-name harmonisation and filtering.

`data/results/` contains outputs produced by the modelling workflow, including coefficient tables, score tables, fitted-value tables, prediction-rate tables, correlation outputs, and source figure files used for the main results.

`data/support/` contains derived diagnostic summaries and auxiliary tables used to interpret or document the modelling results. These include excluded-country records, residual summaries, coefficient-stability summaries, and source figure files for robustness and residual diagnostics.

B Student Visa Application Preparation

The INZ student visa decision data are useful for this analysis partly because a fee-paying student visa application is unlikely to be a casual or spontaneous action. Before an application can be assessed, the applicant normally has to assemble educational, financial, personal, health, and documentary evidence. This does not prove that every application is strong, complete, or equally supported. It does mean that the published decision records represent applications that passed through a substantial preparation process before reaching a formal decision.

This preparation burden matters for interpretation. The observed INZ application counts are not treated as a simple measure of abstract interest in

Table 1. Typical preparation requirements before a fee-paying student visa decision

Preparation area	Examples of required or commonly relevant evidence
Education pathway	Identifying an education provider, submitting documents to that provider, receiving an offer of place, and documenting previous education.
Tuition and insurance	Paying tuition or documenting tuition arrangements, receiving confirmation from the education provider, and arranging medical or travel insurance where required.
Financial capacity	Providing bank statements or other evidence of funds for study and living costs, together with explanations of source of funds and relevant incoming transfers.
Personal and travel history	Providing previous education and work history, explaining gaps in education or employment, and documenting travel history where requested.
Health and character	Providing relevant health information, medical examination material where required, and character-related information required by the visa process.
Home-country ties and translations	Documenting family relationships, property or other ties where relevant, and providing certified translations for documents not originally in English.

New Zealand study. They are better understood as counts of completed applications that were submitted after applicants had already undertaken a costly and document-intensive process. For this reason, variation in approval rates may reflect differences in the practical conditions under which applicants can assemble convincing evidence, not only differences in demand for study.

C Countries Excluded Before Regression

Table 2 reports countries present in the aggregated INZ decision data but not retained in the final regression table. Eight countries were not retained after the reference-data join or missing-value filter. The remaining countries were excluded because they had fewer than 10 total applications across 2022–2025.

Table 2: Countries excluded before regression modelling

Country	Applications	Approval rate	Exclusion reason
Hong Kong	1397	0.981	Not retained after reference join or missing-value filter
Taiwan	1094	0.976	Not retained after reference join or missing-value filter
Macau	69	0.928	Not retained after reference join or missing-value filter
Tuvalu	59	0.983	Not retained after reference join or missing-value filter
Palestine	15	0.533	Not retained after reference join or missing-value filter
Lithuania	10	0.900	Not retained after reference join or missing-value filter
Malawi	10	0.600	Not retained after reference join or missing-value filter
Trinidad and Tobago	10	0.800	Not retained after reference join or missing-value filter
Iceland	9	1.000	Fewer than 10 total applications
Liberia	9	0.000	Fewer than 10 total applications
Namibia	9	0.556	Fewer than 10 total applications
Guatemala	8	0.750	Fewer than 10 total applications
Nauru	8	1.000	Fewer than 10 total applications
Albania	7	0.714	Fewer than 10 total applications
Benin	7	0.286	Fewer than 10 total applications
Estonia	7	1.000	Fewer than 10 total applications
Georgia	7	0.571	Fewer than 10 total applications
Croatia	6	1.000	Fewer than 10 total applications
El Salvador	6	1.000	Fewer than 10 total applications
Guyana	6	0.833	Fewer than 10 total applications
Marshall Islands	6	1.000	Fewer than 10 total applications
South Sudan	6	0.333	Fewer than 10 total applications
St Lucia	6	1.000	Fewer than 10 total applications
Azerbaijan	5	0.800	Fewer than 10 total applications
Gambia	5	0.200	Fewer than 10 total applications
Haiti	5	0.200	Fewer than 10 total applications
Mozambique	5	0.800	Fewer than 10 total applications
Palau	5	1.000	Fewer than 10 total applications
Suriname	5	1.000	Fewer than 10 total applications
Swaziland	5	0.800	Fewer than 10 total applications
Tunisia	5	0.400	Fewer than 10 total applications
Armenia	4	0.500	Fewer than 10 total applications
Bahrain	4	1.000	Fewer than 10 total applications

Table 2 – continued from previous page

Country	Applications	Approval rate	Exclusion reason
Belize	4	1.000	Fewer than 10 total applications
Burundi	4	0.000	Fewer than 10 total applications
Eritrea	4	0.750	Fewer than 10 total applications
Grenada	4	0.750	Fewer than 10 total applications
Macedonia	4	1.000	Fewer than 10 total applications
Mali	4	0.750	Fewer than 10 total applications
Moldova	4	0.750	Fewer than 10 total applications
Serbia	4	0.750	Fewer than 10 total applications
Sierra Leone	4	0.000	Fewer than 10 total applications
Angola	3	0.000	Fewer than 10 total applications
Barbados	3	1.000	Fewer than 10 total applications
Congo	3	0.000	Fewer than 10 total applications
Dominica	3	0.667	Fewer than 10 total applications
Federated States of Micronesia	3	1.000	Fewer than 10 total applications
Liechtenstein	3	1.000	Fewer than 10 total applications
Malta	3	1.000	Fewer than 10 total applications
Slovenia	3	1.000	Fewer than 10 total applications
Somalia	3	0.000	Fewer than 10 total applications
St Kitts - Nevis	3	0.667	Fewer than 10 total applications
St Vincent and the Grenadines	3	1.000	Fewer than 10 total applications
Tajikistan	3	0.667	Fewer than 10 total applications
Australia	2	1.000	Fewer than 10 total applications
Bermuda	2	0.500	Fewer than 10 total applications
Guinea	2	0.000	Fewer than 10 total applications
Senegal	2	0.000	Fewer than 10 total applications
Seychelles	2	1.000	Fewer than 10 total applications
Bahamas	1	1.000	Fewer than 10 total applications
Bosnia & Herzegovina	1	1.000	Fewer than 10 total applications
Cape Verde	1	1.000	Fewer than 10 total applications
Cuba	1	1.000	Fewer than 10 total applications
Djibouti	1	0.000	Fewer than 10 total applications
Madagascar	1	0.000	Fewer than 10 total applications
Nicaragua	1	1.000	Fewer than 10 total applications
Niger	1	0.000	Fewer than 10 total applications
North Korea	1	0.000	Fewer than 10 total applications
Togo	1	0.000	Fewer than 10 total applications

D Full Correlation Table

Table 3 reports all pairwise Pearson correlations among approval rate and the numerical country-level indicators used in the correlation output.

Table 3: Full pairwise correlation table

First variable	Second variable	r	p	df
Approval rate	Under-five mortality	-0.746	1.24e-30	163
Approval rate	Maternal mortality	-0.681	7.97e-24	163
Approval rate	Physician density	0.552	1.56e-14	163
Approval rate	DTP3 immunization coverage	0.371	9e-7	163
Approval rate	GDP per capita	0.501	7.42e-12	163
Approval rate	Life expectancy	0.715	4.19e-27	163
Approval rate	Internet users	0.633	7.62e-20	163
Approval rate	Urban population	0.340	8.12e-6	163
Approval rate	GNI per capita	0.520	8.18e-13	163

Table 3 – continued from previous page

First variable	Second variable	<i>r</i>	<i>p</i>	df
Under-five mortality	Maternal mortality	0.881	9.52e-55	163
Under-five mortality	Physician density	-0.652	2.36e-21	163
Under-five mortality	DTP3 immunization coverage	-0.487	3.39e-11	163
Under-five mortality	GDP per capita	-0.488	3.12e-11	163
Under-five mortality	Life expectancy	-0.888	7.59e-57	163
Under-five mortality	Internet users	-0.770	1.19e-33	163
Under-five mortality	Urban population	-0.521	7.37e-13	163
Under-five mortality	GNI per capita	-0.504	5.16e-12	163
Maternal mortality	Physician density	-0.569	1.48e-15	163
Maternal mortality	DTP3 immunization coverage	-0.455	8.61e-10	163
Maternal mortality	GDP per capita	-0.398	1.21e-7	163
Maternal mortality	Life expectancy	-0.813	3.93e-40	163
Maternal mortality	Internet users	-0.723	5.36e-28	163
Maternal mortality	Urban population	-0.448	1.63e-9	163
Maternal mortality	GNI per capita	-0.413	3.66e-8	163
Physician density	DTP3 immunization coverage	0.382	4.27e-7	163
Physician density	GDP per capita	0.554	1.2e-14	163
Physician density	Life expectancy	0.717	2.62e-27	163
Physician density	Internet users	0.670	7e-23	163
Physician density	Urban population	0.559	6.23e-15	163
Physician density	GNI per capita	0.580	3.05e-16	163
DTP3 immunization coverage	GDP per capita	0.352	3.52e-6	163
DTP3 immunization coverage	Life expectancy	0.500	8.48e-12	163
DTP3 immunization coverage	Internet users	0.501	7.14e-12	163
DTP3 immunization coverage	Urban population	0.284	2.14e-4	163
DTP3 immunization coverage	GNI per capita	0.363	1.6e-6	163
GDP per capita	Life expectancy	0.671	6.45e-23	163
GDP per capita	Internet users	0.546	3.16e-14	163
GDP per capita	Urban population	0.548	2.67e-14	163
GDP per capita	GNI per capita	0.980	7.41e-116	163
Life expectancy	Internet users	0.804	1.32e-38	163
Life expectancy	Urban population	0.657	9.55e-22	163
Life expectancy	GNI per capita	0.694	4.51e-25	163
Internet users	Urban population	0.724	4e-28	163
Internet users	GNI per capita	0.566	2.33e-15	163
Urban population	GNI per capita	0.573	8.94e-16	163

The following tables report the full coefficient and p-value outputs for the four regression specifications discussed in Section 5.3.

Table 4: Full coefficient table for the UN M49 regional specification

Term	Coefficient	<i>p</i>
Urban_population_pct	-0.2795	0.0053
UN M49=Northern Africa	-0.2537	0.0109
Life_expectancy_at_birth_years	0.4992	0.0128
WB_FY2026_Lending_Category=Missing	0.2425	0.0156
UN M49=Middle Africa	-0.2749	0.0219
UN M49=Southern Asia	-0.2100	0.0336
Intercept	0.4674	0.0492
WB_FY2026_Income_Group=Low income	-0.1980	0.0714
UN M49=Melanesia	0.2055	0.0861
UN M49=Central Asia	-0.1989	0.0902
UN M49=Western Asia	-0.1674	0.0904
WB_FY2026_Lending_Category=IBRD	0.1036	0.1439
UN M49=South-eastern Asia	0.1074	0.2368
WB_FY2026_Lending_Category=IDA	0.0600	0.3012

Table 4 – continued from previous page

Term	Coefficient	<i>p</i>
UN M49=Micronesia	0.1634	0.3124
Internet_users_pct_population	0.1136	0.3553
UN M49=Southern Europe	-0.0925	0.4785
UN M49=Polynesia	0.0970	0.4848
GDP_per_capita_current_USD	-0.1922	0.4888
UN M49=Northern Europe	-0.0755	0.5001
GNI_per_capita_atlas_current_USD	0.1768	0.5223
UN M49=Central America	0.0613	0.5712
WB_FY2026_Income_Group=Lower middle income	-0.0389	0.5992
UN M49=Eastern Africa	-0.0478	0.6506
UN M49=South America	0.0438	0.6554
WHO_DTP3_Immunization_Coverage_pct	-0.0355	0.6678
UN M49=Western Europe	-0.0315	0.7865
WB_FY2026_Income_Group=Missing	-0.0290	0.7943
UN M49=Northern America	-0.0335	0.7951
WB_FY2026_Income_Group=Upper middle income	-0.0148	0.7967
UN M49=Eastern Asia	-0.0261	0.8053
UN M49=Eastern Europe	0.0192	0.8582
UN M49=Southern Africa	0.0204	0.8628
WHO_Physician_Density_per_10000_population	-0.0140	0.9018
WHO_Under5_Mortality_per_1000_live_births	0.0245	0.9164
UN M49=Western Africa	-0.0105	0.9384
WHO_Maternal_Mortality_per_100000_live_births	-0.0081	0.9689

Table 5: Full coefficient table for the World Bank regional specification

Term	Coefficient	<i>p</i>
WB_FY2026_Region=Middle East, North Africa, Afghanistan & Pakistan	-0.2695	8.194e-8
Intercept	0.9166	5.844e-7
WB_FY2026_Income_Group=Low income	-0.4067	5.477e-5
WB_FY2026_Region=Europe & Central Asia	-0.1846	3.689e-4
WB_FY2026_Region=South Asia	-0.2322	5.548e-4
WB_FY2026_Region=Sub-Saharan Africa	-0.1561	0.0042
Urban_population_pct	-0.2545	0.0061
WB_FY2026_Lending_Category=Missing	0.2244	0.0171
WB_FY2026_Income_Group=Lower middle income	-0.1355	0.0461
WB_FY2026_Income_Group=Upper middle income	-0.0914	0.0889
WB_FY2026_Lending_Category=IBRD	0.1114	0.0918
WB_FY2026_Region=North America	-0.1633	0.1001
Life_expectancy_at_birth_years	0.2595	0.1672
WB_FY2026_Region=Latin America & Caribbean	-0.0644	0.2041
WHO_DTP3_Immunization_Coverage_pct	-0.0970	0.2097
WB_FY2026_Lending_Category=IDA	0.0692	0.2267
WB_FY2026_Income_Group=Missing	-0.1243	0.2511
WHO_Under5_Mortality_per_1000_live_births	-0.1944	0.3301
GNI_per_capita_atlas_current_USD	0.1541	0.5663
GDP_per_capita_current_USD	-0.1087	0.7076
Internet_users_pct_population	0.0271	0.7942
WHO_Maternal_Mortality_per_100000_live_births	-0.0367	0.8419
WHO_Physician_Density_per_10000_population	-0.0041	0.9680

Table 6: Full coefficient table for the WHO regional specification

Term	Coefficient	<i>p</i>
Intercept	0.8794	3.670e-6
WB_FY2026_Income_Group=Low income	-0.4423	4.232e-5
WHO_Region=Western Pacific Region	0.1980	9.745e-4
WB_FY2026_Income_Group=Lower middle income	-0.1780	0.0127
Urban_population_pct	-0.2370	0.0132
WB_FY2026_Lending_Category=Missing	0.2311	0.0198
WB_FY2026_Income_Group=Upper middle income	-0.1039	0.0680
WHO_Region=Region of the Americas	0.1219	0.0691
WB_FY2026_Lending_Category=IBRD	0.1173	0.0952
WHO_DTP3_Immunization_Coverage_pct	-0.1333	0.1146
WHO_Region=South-East Asia Region	0.0849	0.1933
WB_FY2026_Income_Group=Missing	-0.1484	0.1958
WHO_Under5_Mortality_per_1000_live_births	-0.2102	0.3220
WHO_Region=Eastern Mediterranean Region	-0.0545	0.3817
WB_FY2026_Lending_Category=IDA	0.0481	0.4213
Life_expectancy_at_birth_years	0.1354	0.4701
GNI_per_capita_atlas_current_USD	0.1327	0.6318
WHO_Region=European Region	0.0319	0.6626
WHO_Physician_Density_per_10000_population	-0.0393	0.7146
GDP_per_capita_current_USD	-0.0843	0.7816
WHO_Maternal_Mortality_per_100000_live_births	-0.0494	0.7995
Internet_users_pct_population	-0.0107	0.9230

Table 7: Full coefficient table for the no-region specification

Term	Coefficient	<i>p</i>
Intercept	1.1160	3.208e-8
WB_FY2026_Income_Group=Low income	-0.6087	2.500e-7
WB_FY2026_Income_Group=Lower middle income	-0.2242	0.0047
Urban_population_pct	-0.2635	0.0081
WHO_DTP3_Immunization_Coverage_pct	-0.2036	0.0152
WHO_Under5_Mortality_per_1000_live_births	-0.4155	0.0758
WB_FY2026_Income_Group=Upper middle income	-0.1059	0.0958
WB_FY2026_Income_Group=Missing	-0.2087	0.1085
WB_FY2026_Lending_Category=Missing	0.1644	0.1301
WB_FY2026_Lending_Category=IDA	0.0833	0.2166
WB_FY2026_Lending_Category=IBRD	0.0930	0.2363
WHO_Physician_Density_per_10000_population	-0.1130	0.2560
GNI_per_capita_atlas_current_USD	0.2416	0.4372
Life_expectancy_at_birth_years	0.0871	0.6543
GDP_per_capita_current_USD	-0.1465	0.6644
Internet_users_pct_population	0.0134	0.9087
WHO_Maternal_Mortality_per_100000_live_births	0.0182	0.9329

E Coefficient Stability Summary

Table 8 reports the cross-specification coefficient stability summary derived from the four coefficient tables. The table gives the number of model specifications in which each term appears, the number of those specifications where $p < 0.05$, and the minimum and maximum p-values observed across the specifications where the term is present.

Table 8: Coefficient stability summary across regression specifications

Term	Type	Models	Sig.	Min. p	Max. p	Stability
Urban_population_pct WB_FY2026_Income_Group=	Variable	4	4	0.0053	0.0132	stable significant
Low income WB_FY2026_Income_Group=	Variable	4	3	$2.5e-7$	0.0714	partly stable
Lower middle income WB_FY2026_Lending_Category=	Variable	4	3	0.0047	0.5992	partly stable
Missing Life_expectancy_at_birth_years	Variable	4	3	0.0156	0.1301	partly stable
WHO_DTP3_Immunization _Coverage_pct	Variable	4	1	0.0128	0.6543	specification dependent
WB_FY2026_Income_Group=	Variable	4	0	0.0152	0.6678	specification dependent
Upper middle income WHO_Under5_Mortality	Variable	4	0	0.0680	0.7967	not significant
_per_1000_live_births WB_FY2026_Lending_Category=	Variable	4	0	0.0758	0.9164	not significant
IBRD WB_FY2026_Income_Group=	Variable	4	0	0.0918	0.2363	not significant
Missing WB_FY2026_Lending_Category=	Variable	4	0	0.1085	0.7943	not significant
IDA WHO_Physician_Density	Variable	4	0	0.2166	0.4213	not significant
_per_10000_population	Variable	4	0	0.2560	0.9680	not significant
Internet_users_pct_population	Variable	4	0	0.3553	0.9230	not significant
GNI_per_capita_Atlas_current_USD	Variable	4	0	0.4372	0.6318	not significant
GDP_per_capita_current_USD	Variable	4	0	0.4888	0.7816	not significant
WHO_Maternal_Mortality _per_100000_live_births	Variable	4	0	0.7995	0.9689	not significant
WB_FY2026_Region=Middle East, North Africa, Afghanistan & Pakistan WB_FY2026_Region=Europe & Central Asia	Region	1	1	$8.19e-8$	$8.19e-8$	stable significant
WB_FY2026_Region=South Asia	Region	1	1	$3.69e-4$	$3.69e-4$	stable significant
WHO_Region=Western Pacific Region	Region	1	1	$5.55e-4$	$5.55e-4$	stable significant
WB_FY2026_Region=Sub-Saharan Africa	Region	1	1	$9.75e-4$	$9.75e-4$	stable significant
UN M49=Northern Africa	Region	1	1	0.0042	0.0042	stable significant
UN M49=Middle Africa	Region	1	1	0.0109	0.0109	stable significant
UN M49=Southern Asia	Region	1	1	0.0219	0.0219	stable significant
WHO_Region=Region of the Americas	Region	1	1	0.0336	0.0336	stable significant
UN M49=Melanesia	Region	1	0	0.0691	0.0691	not significant
UN M49=Central Asia	Region	1	0	0.0861	0.0861	not significant
UN M49=Western Asia	Region	1	0	0.0902	0.0902	not significant
WB_FY2026_Region=North America	Region	1	0	0.0904	0.0904	not significant
WHO_Region=South-East Asia Region	Region	1	0	0.1001	0.1001	not significant
WB_FY2026_Region=Latin America & Caribbean	Region	1	0	0.1933	0.1933	not significant
UN M49=South-eastern Asia	Region	1	0	0.2041	0.2041	not significant
UN M49=Micronesia	Region	1	0	0.2368	0.2368	not significant
WHO_Region=Eastern Mediterranean Re- gion	Region	1	0	0.3124	0.3124	not significant
UN M49=Southern Europe	Region	1	0	0.3817	0.3817	not significant
UN M49=Polynesia	Region	1	0	0.4785	0.4785	not significant
UN M49=Northern Europe	Region	1	0	0.4848	0.4848	not significant
UN M49=Central America	Region	1	0	0.5001	0.5001	not significant
UN M49=Eastern Africa	Region	1	0	0.5712	0.5712	not significant
UN M49=South America	Region	1	0	0.6506	0.6506	not significant
WHO_Region=European Region	Region	1	0	0.6554	0.6554	not significant
UN M49=Western Europe	Region	1	0	0.6626	0.6626	not significant
UN M49=Northern America	Region	1	0	0.7865	0.7865	not significant
UN M49=Eastern Asia	Region	1	0	0.7951	0.7951	not significant
UN M49=Eastern Europe	Region	1	0	0.8053	0.8053	not significant
UN M49=Southern Africa	Region	1	0	0.8582	0.8582	not significant
UN M49=Western Africa	Region	1	0	0.8628	0.8628	not significant
	Region	1	0	0.9384	0.9384	not significant

F Relative Residual Tables

Tables 9 and 10 report the largest positive and negative relative residuals by model specification. Each entry gives the country, **Prediction_rate**, and residual delta. The tables are derived from the model-specific fitted-value outputs.

Table 9: Largest positive relative residuals by model specification

UN M49	World Bank	WHO	No region
Democratic Republic of Congo rate=2.375 delta=0.109	Nigeria rate=1.940 delta=0.204	Rwanda rate=1.852 delta=0.241	Rwanda rate=2.191 delta=0.285
Sudan rate=1.551 delta=0.127	Rwanda rate=1.601 delta=0.197	Nigeria rate=1.736 delta=0.178	Timor Leste rate=1.804 delta=0.437
Nigeria rate=1.392 delta=0.118	Timor Leste rate=1.410 delta=0.285	Timor Leste rate=1.642 delta=0.384	Uganda rate=1.774 delta=0.200
Timor Leste rate=1.346 delta=0.252	Uganda rate=1.364 delta=0.122	Uganda rate=1.508 delta=0.154	Nigeria rate=1.659 delta=0.167
Kazakhstan rate=1.269 delta=0.161	Botswana rate=1.242 delta=0.146	Jordan rate=1.247 delta=0.118	Kiribati rate=1.624 delta=0.299
Jamaica rate=1.198 delta=0.151	Ukraine rate=1.233 delta=0.162	Botswana rate=1.245 delta=0.148	Laos rate=1.360 delta=0.254
Zambia rate=1.195 delta=0.103	Lebanon rate=1.213 delta=0.122	Kiribati rate=1.240 delta=0.151	Solomon Islands rate=1.300 delta=0.222
Lebanon rate=1.184 delta=0.108	Kiribati rate=1.202 delta=0.131	Zambia rate=1.238 delta=0.121	Malaysia rate=1.287 delta=0.218
China rate=1.175 delta=0.144	Laos rate=1.196 delta=0.157	Laos rate=1.207 delta=0.165	Fiji rate=1.282 delta=0.175
Bolivia rate=1.171 delta=0.121	Kuwait rate=1.185 delta=0.154	Kuwait rate=1.203 delta=0.166	Brazil rate=1.276 delta=0.193
Laos rate=1.169 delta=0.139	Jordan rate=1.179 delta=0.090	Indonesia rate=1.198 delta=0.157	Cambodia rate=1.270 delta=0.195
Sri Lanka rate=1.164 delta=0.109	Brazil rate=1.176 delta=0.134	Lebanon rate=1.198 delta=0.115	China rate=1.240 delta=0.187
Maldives rate=1.154 delta=0.110	Jamaica rate=1.171 delta=0.134	Ukraine rate=1.198 delta=0.141	Jamaica rate=1.206 delta=0.157
Oman rate=1.146 delta=0.122	Poland rate=1.166 delta=0.143	Brazil rate=1.192 delta=0.143	Philippines rate=1.198 delta=0.138
Rwanda rate=1.144 delta=0.066	Zambia rate=1.165 delta=0.090	Solomon Islands rate=1.168 delta=0.139	Mexico rate=1.196 delta=0.151

Table 10: Largest negative relative residuals by model specification

UN M49	World Bank	WHO	No region
Chad rate=0.000 delta=-0.032	Chad rate=0.000 delta=-0.133	Chad rate=0.000 delta=-0.151	Chad rate=0.000 delta=-0.073
Afghanistan rate=0.495 delta=-0.144	Cameroon rate=0.342 delta=-0.266	Cameroon rate=0.347 delta=-0.259	Cameroon rate=0.334 delta=-0.275
Uzbekistan rate=0.522 delta=-0.229	Uzbekistan rate=0.443 delta=-0.314	Uzbekistan rate=0.450 delta=-0.306	Uzbekistan rate=0.423 delta=-0.342

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UN M49	World Bank	WHO	No region
Turkey rate=0.597 delta=-0.237	Turkey rate=0.550 delta=-0.288	Afghanistan rate=0.517 delta=-0.132	Afghanistan rate=0.447 delta=-0.174
Myanmar rate=0.626 delta=-0.294	Afghanistan rate=0.615 delta=-0.088	Turkey rate=0.541 delta=-0.298	Turkey rate=0.516 delta=-0.330
Cameroon rate=0.642 delta=-0.077	Myanmar rate=0.619 delta=-0.304	Algeria rate=0.643 delta=-0.231	Libya rate=0.571 delta=-0.273
Yemen rate=0.704 delta=-0.105	Tanzania rate=0.710 delta=-0.173	Nepal rate=0.683 delta=-0.180	Algeria rate=0.587 delta=-0.293
Tanzania rate=0.719 delta=-0.166	Algeria rate=0.726 delta=-0.157	Libya rate=0.688 delta=-0.165	Yemen rate=0.610 delta=-0.160
Algeria rate=0.794 delta=-0.108	Democratic Republic of Congo rate=0.727 delta=-0.070	Yemen rate=0.718 delta=-0.098	Tanzania rate=0.645 delta=-0.233
Ghana rate=0.794 delta=-0.118	Mongolia rate=0.727 delta=-0.227	Myanmar rate=0.719 delta=-0.193	Iraq rate=0.650 delta=-0.256
Ecuador rate=0.802 delta=-0.176	Libya rate=0.749 delta=-0.122	Mongolia rate=0.726 delta=-0.228	Nepal rate=0.693 delta=-0.172
Dominican Republic rate=0.803 delta=-0.151	Dominican Republic rate=0.810 delta=-0.145	Democratic Republic of Congo rate=0.727 delta=-0.070	Myanmar rate=0.698 delta=-0.213
Iraq rate=0.835 delta=-0.094	Ecuador rate=0.814 delta=-0.163	Tanzania rate=0.759 delta=-0.135	Iran rate=0.730 delta=-0.197
Vietnam rate=0.851 delta=-0.143	Nepal rate=0.827 delta=-0.081	India rate=0.765 delta=-0.173	Democratic Republic of Congo rate=0.789 delta=-0.050
Syria rate=0.853 delta=-0.042	Yemen rate=0.842 delta=-0.047	Iraq rate=0.797 delta=-0.121	Zimbabwe rate=0.790 delta=-0.106

G Prediction Table

Table 11 reports the observed approval rate and the fitted approval rates from the four regression specifications for each country in the modelling table. The table is derived from the fitted-value outputs for the UN M49, World Bank, WHO, and no-region specifications.

Table 11: Observed and fitted approval rates by regression specification

Country	Observed	UN M49	WB	WHO	No region
Afghanistan	0.141	0.285	0.229	0.273	0.315
Algeria	0.417	0.525	0.574	0.648	0.710
Argentina	0.858	0.803	0.797	0.775	0.721
Austria	1.000	1.038	1.016	1.006	1.022
Bangladesh	0.558	0.587	0.605	0.678	0.669
Belarus	0.750	0.759	0.664	0.671	0.703
Belgium	0.995	0.982	0.946	0.924	0.909
Bhutan	0.632	0.572	0.588	0.656	0.669

Table 11 – continued from previous page

Country	Observed	UN M49	WB	WHO	No region
Bolivia	0.826	0.705	0.728	0.740	0.696
Botswana	0.750	0.686	0.604	0.602	0.657
Brazil	0.892	0.781	0.758	0.748	0.699
Brunei Darussalam	1.000	1.058	1.086	1.093	0.958
Bulgaria	0.833	0.803	0.783	0.805	0.841
Cambodia	0.916	0.834	0.818	0.796	0.721
Cameroon	0.138	0.215	0.403	0.397	0.413
Canada	0.986	1.002	0.986	1.056	0.997
Chad	0.000	0.032	0.133	0.151	0.073
Chile	0.901	0.891	0.908	0.879	0.824
China	0.966	0.822	0.919	0.899	0.779
Colombia	0.748	0.836	0.806	0.790	0.741
Costa Rica	0.875	0.921	0.918	0.892	0.834
Czech Republic	0.990	1.032	0.968	0.976	0.990
Democratic Republic of Congo	0.188	0.079	0.258	0.258	0.238
Denmark	0.997	0.955	0.963	0.949	0.973
Dominican Republic	0.615	0.767	0.760	0.752	0.687
Ecuador	0.714	0.890	0.878	0.873	0.851
Egypt	0.561	0.526	0.588	0.595	0.694
Ethiopia	0.641	0.645	0.651	0.637	0.713
Fiji	0.795	0.783	0.687	0.721	0.620
Finland	0.912	0.988	0.997	0.992	1.022
France	0.978	1.018	0.971	0.965	0.981
Germany	0.996	0.982	0.963	0.963	0.984
Ghana	0.455	0.573	0.516	0.464	0.555
Great Britain	0.984	0.941	0.953	0.955	0.979
Greece	0.917	0.917	0.942	0.926	0.893
Honduras	0.727	0.756	0.746	0.717	0.721
Hungary	0.982	0.983	0.919	0.934	0.942
India	0.566	0.598	0.632	0.739	0.686
Indonesia	0.950	0.859	0.870	0.793	0.820
Iran	0.531	0.590	0.594	0.603	0.727
Iraq	0.476	0.570	0.559	0.597	0.732
Ireland	0.981	1.004	1.027	1.019	1.049
Israel	0.865	0.843	0.854	0.931	0.940
Italy	0.995	0.999	1.007	0.992	1.000
Jamaica	0.917	0.765	0.783	0.787	0.760
Japan	0.996	0.994	1.119	1.096	0.942
Jordan	0.595	0.570	0.505	0.477	0.543
Kazakhstan	0.762	0.601	0.705	0.712	0.748
Kenya	0.464	0.487	0.494	0.482	0.547
Kiribati	0.779	0.779	0.648	0.628	0.479
Kuwait	0.984	0.865	0.830	0.818	0.914
Kyrgyzstan	0.667	0.599	0.683	0.655	0.740
Laos	0.960	0.821	0.803	0.795	0.706
Latvia	0.917	0.892	0.928	0.945	0.961
Lebanon	0.696	0.588	0.574	0.581	0.689
Libya	0.364	0.406	0.485	0.528	0.637
Luxembourg	1.000	0.923	0.916	0.916	0.936
Malaysia	0.976	0.910	0.878	0.869	0.758
Maldives	0.827	0.717	0.737	0.797	0.858
Mauritius	0.865	0.804	0.790	0.757	0.855
Mexico	0.923	0.818	0.803	0.802	0.772
Mongolia	0.605	0.698	0.831	0.832	0.689
Morocco	0.529	0.541	0.566	0.555	0.652
Myanmar	0.493	0.786	0.796	0.685	0.706
Nepal	0.388	0.391	0.469	0.568	0.560
Netherlands	0.990	0.959	0.933	0.931	0.955

Table 11 – continued from previous page

Country	Observed	UN M49	WB	WHO	No region
Nigeria	0.420	0.302	0.216	0.242	0.253
Norway	0.983	1.009	1.014	0.991	1.031
Oman	0.956	0.835	0.827	0.833	0.924
Pakistan	0.391	0.402	0.379	0.399	0.458
Panama	0.919	0.949	0.984	0.988	0.974
Papua New Guinea	0.863	0.847	0.803	0.858	0.769
Paraguay	0.864	0.806	0.803	0.798	0.744
Peru	0.716	0.826	0.796	0.789	0.751
Philippines	0.838	0.822	0.824	0.840	0.700
Poland	1.000	0.902	0.857	0.865	0.913
Portugal	0.963	0.987	0.998	0.977	0.956
Romania	0.913	0.909	0.882	0.906	0.976
Russia	0.693	0.775	0.763	0.783	0.818
Rwanda	0.524	0.458	0.327	0.283	0.239
Samoa	0.955	0.880	0.940	0.957	0.927
Saudi Arabia	0.772	0.808	0.826	0.831	0.928
Singapore	0.964	1.091	1.111	1.091	0.956
Slovakia	1.000	1.066	0.987	0.993	1.001
Solomon Islands	0.963	0.912	0.834	0.825	0.741
South Africa	0.621	0.685	0.639	0.652	0.744
South Korea	0.985	1.037	1.153	1.126	0.981
Spain	0.997	0.969	0.972	0.956	0.957
Sri Lanka	0.778	0.668	0.716	0.777	0.806
Sudan	0.357	0.230	0.370	0.317	0.373
Sweden	0.988	0.973	0.967	0.949	0.962
Switzerland	0.980	1.037	0.999	0.978	1.008
Syria	0.243	0.285	0.210	0.227	0.255
Tanzania	0.424	0.590	0.597	0.559	0.657
Thailand	0.925	0.949	0.916	0.805	0.827
Timor Leste	0.981	0.729	0.696	0.598	0.544
Tonga	0.863	0.937	0.944	0.928	0.895
Turkey	0.352	0.589	0.640	0.650	0.682
Uganda	0.458	0.437	0.336	0.304	0.258
Ukraine	0.857	0.789	0.695	0.716	0.760
United Arab Emirates	1.000	0.881	0.875	0.865	0.974
United States of America	0.984	0.967	0.983	1.058	1.011
Uruguay	0.828	0.812	0.866	0.845	0.783
Uzbekistan	0.250	0.479	0.564	0.556	0.592
Vanuatu	0.878	0.957	0.901	0.900	0.843
Venezuela	0.718	0.714	0.708	0.722	0.646
Vietnam	0.818	0.961	0.893	0.863	0.701
Yemen	0.250	0.355	0.297	0.348	0.410
Zambia	0.632	0.528	0.542	0.510	0.567
Zimbabwe	0.400	0.458	0.459	0.449	0.506

References

1. Kaplan, V.: Inz visa decisions research repository. GitHub repository (2026), <https://github.com/vitalii-kaplan/inz-visa-decisions>, retrieved May 13, 2026